

Day -2 : Power Supply System Theory

Task -1: real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply),

1. Devices that use SMPS (Switched Mode Power Supply)

- **Desktop PC Power Supply (PSU):** It converts high-voltage AC wall power into steady, regulated low-voltage DC power required by sensitive computer components.
- **Mobile Phone Charger:** It converts household AC power into low-voltage DC efficiently using a lightweight, compact design.
- **LED TV / Monitor:** It provides precise, energy-efficient DC voltages to power the screen's backlighting and internal electronics without overheating.

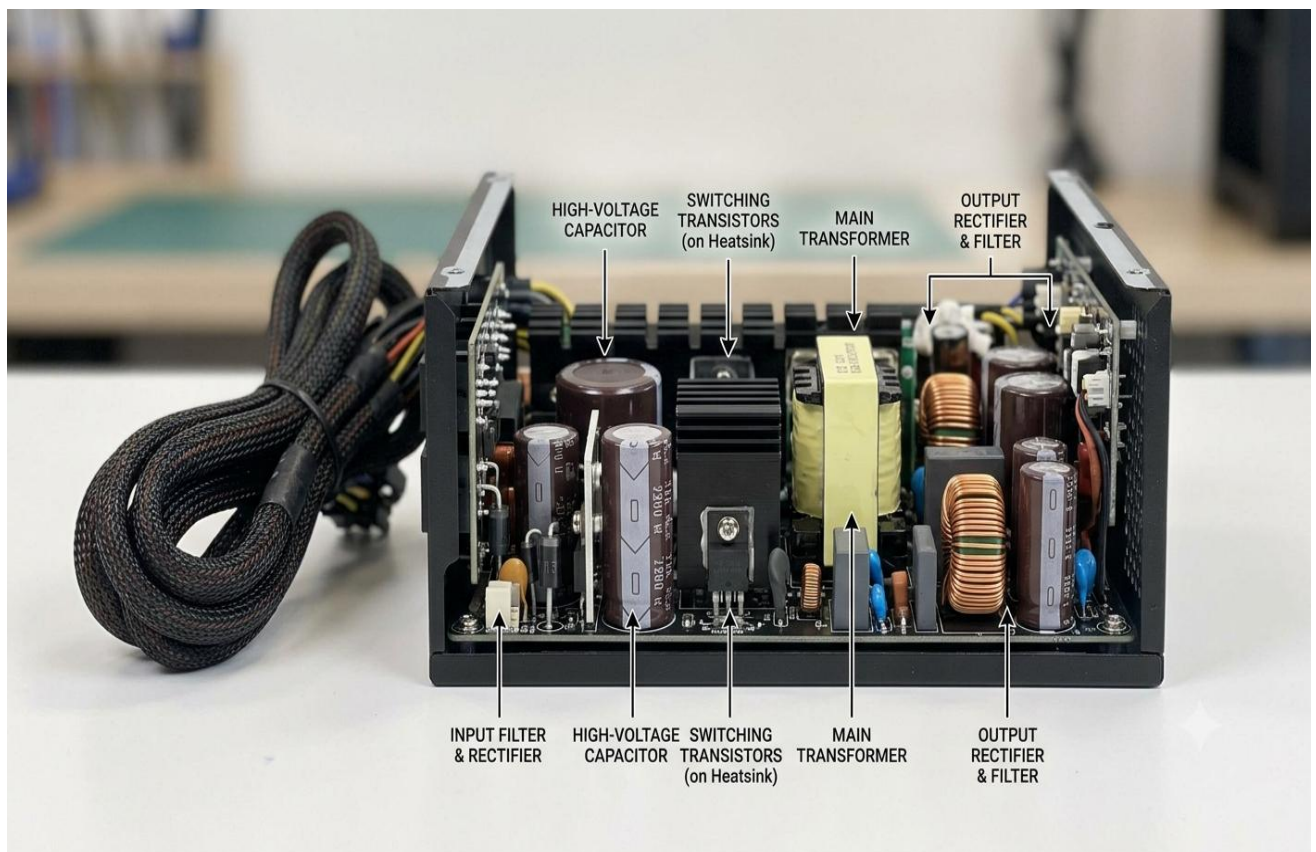
2. Devices that use UPS (Uninterruptible Power Supply)

- **Enterprise Network Server:** It requires instant battery backup to prevent sudden data

loss and system corruption during unexpected power cuts.

- **College Wi-Fi Router / Network Switch:** It stays powered on during electricity outages so students and staff maintain continuous internet access.

Task -2 : Labelled Diagram of SMPS

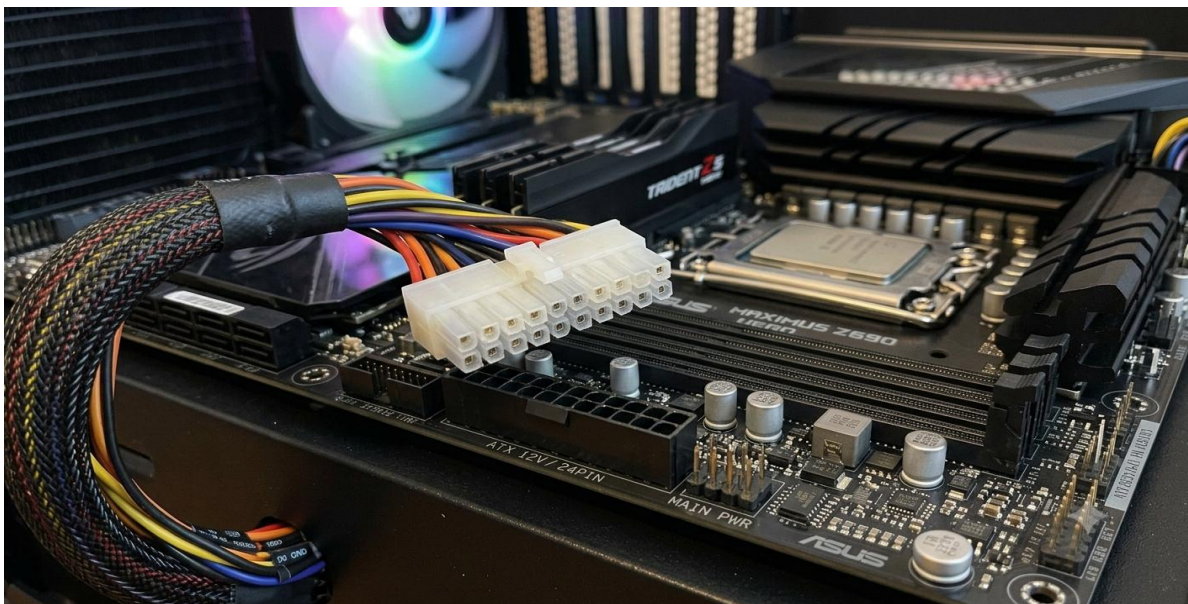


Task -3 : Diagram of ATX -24 pin ,SATA ,Molex connector and PCIe 6,8 Pin

SATA Power Connector



ATX 20+4 Pin Diagram



Molex Connector



PCIe 6/8 Pin Connector



Task -4 : building a pc

1. Main Motherboard Power

- **Cable: ATX 24-Pin (the biggest cable coming from the power supply).**
- **Where it goes: Into the long 24-pin socket on the right side of the motherboard.**
- **Action: Push it in until the plastic tab clicks.**

2. CPU Power

- **Cable: EPS / CPU Cable (CPU, usually an 8-pin or 4+4 pin).**
- **Where it goes: Top-left corner of the motherboard.**
- **Action: Line up the clip and press until it clicks. *(Don't confuse this with the graphics card cable!)***

3. Graphics Card (GPU) Power

- **Cable: PCIe Cable (6+2 pin) or 12VHPWR cable for newer cards.**
- **Where it goes: The side or top of your graphics card.**
- **Action: Plug it in firmly until there is no gap and the latch clicks.**

4. Storage Drives

- **M.2 SSDs: No cables needed. They plug directly into the motherboard.**
- **SATA SSDs / Hard Drives: Use the flat, L-shaped SATA Power Cable. Plug it into the back of the drive.**